

1A)

To construct a different ciphertext that also decrypts to the same M , we can simply multiply the second component h by a fixed group element y that has order 2 in G . Since $y^2 = 1$, we have:

$$(y * h) = y * (g^x * M^r \bmod p) = y * g^x * M^r \bmod p$$

the decryption of the original ciphertext (g, h) is given by:

$$M = h / (g^x)^r \bmod p = g^x * M^r / g^{(xr)} \bmod p = M^{(r*x)} \bmod p$$

Similarly, the decryption of the new ciphertext $(g, y * h)$ is given by:

$$M' = (y * h) / (g^x)^r \bmod p = y * g^x * M^r / g^{(xr)} \bmod p = y * M^{(r*x)} \bmod p$$

Since $y^2 = 1$, we can substitute $y^2 = 1$ into the equation for M' to obtain:

$$M' = y * M^{(rx)} \bmod p = y^2 * M^{(rx)} \bmod p = M^{(r*x)} \bmod p = M$$

Therefore, we have constructed a different ElGamal ciphertext that also decrypts to the same plaintext M .

1B)

Let's assume that we have been given two ElGamal ciphertexts:

$$(c1, d1) = (g^{k1}, M1 * h^{k1} \bmod p)$$

$$(c2, d2) = (g^{k2}, M2 * h^{k2} \bmod p)$$

where g is the generator of the group G , h is the public key of the recipient, and p is the prime order of the group G .

To construct a ciphertext that decrypts to their product $M1 * M2$, we can simply multiply these two ciphertexts:

$$(c1 * c2, d1 * d2) = (g^{k1} * g^{k2}, M1 * M2 * h^{k1} * h^{k2} \bmod p)$$

$$= (g^{(k1 + k2)}, M1 * M2 * h^{(k1 + k2)} \bmod p)$$

The resulting ciphertext is an ElGamal ciphertext for $M1 * M2$ with the same public key h . To decrypt this ciphertext, the recipient can use their secret key a to compute:

$$(d1 * d2) / (c1 * c2)^a = (M1 * M2 * h^{(k1 + k2)} \bmod p) / (g^{(k1 + k2)})^a \bmod p$$

$$= (M1 * M2 * h^{(k1 + k2 - a*(k1 + k2))} \bmod p)$$

$$= (M1 * M2 * h^{(k1 + k2 - k1a - k2a)} \bmod p)$$

$$= (M1 * M2 * h^{(-a*(k1 + k2))} \bmod p)$$

Since the recipient knows the values of $k1$, $k2$, and a , they can compute the decryption of the product $M1 \cdot M2$ from the ciphertext $(c1 * c2, d1 * d2)$.

2A)

If $B1 = B2$, we can exploit the homomorphic property of the ElGamal encryption scheme, which states that multiplying two ciphertexts $(c1, c2)$ and $(d1, d2)$ encrypted under the same public key A results in a new ciphertext $(c1d1, c2d2)$ that encrypts the sum of the two plaintexts.

If we multiply the two ciphertexts $(B1, C1)$ and $(B2, C2)$ together, we get $(B1B2, C1C2)$, which encrypts the product of the plaintexts $M1 * M2$.

We are not given the value of $M1M2$, but only the fact that $B1 = B2$. This means that we can solve for the unknown value $M1M2$ by computing the discrete logarithm of $B1$ and then dividing the discrete logarithm of $B1$ by the discrete logarithm of $B2$. This gives us the discrete logarithm of $M1M2$, which we can then use to recover the value of $M1M2$.

Once we have the value of $M1 * M2$, we can either try to factor it to recover the individual values of $M1$ and $M2$, or we can use other techniques to infer information about $M1$ and $M2$ based on the knowledge of their product.

2B)

If you observe that $B1 = g \cdot B2$ in ElGamal encryption, it implies that the two ciphertexts share the same random value k used for encryption. Specifically, we can write:

$$B1 = g^k \bmod p$$

$$B2 = g^k \bmod p$$

where p is the order of the cyclic group generated by g . This means that the two ciphertexts were encrypted using the same random value k .

Now, let's denote the private key corresponding to the public key A as a , i.e., $A = g^a \bmod p$. Then, we can compute the shared secret key as follows:

$$s = B2^a \bmod p$$

Since we know $B1 = g \cdot B2$, we can substitute $B2$ in the above equation to get:

$$s = (g \cdot B1)^a \bmod p$$

$$= g^a \cdot B1^a \bmod p$$

Using the shared secret key s , we can decrypt both ciphertexts and obtain the corresponding plaintexts:

$$M1 = C1 / s$$

$$M2 = C2 / s$$

Therefore, if we observe that $B_1 = g \cdot B_2$, we can compute the shared secret key s and use it to decrypt both ciphertexts. However, we cannot infer any information about the plaintexts M_1 and M_2 themselves without further knowledge or assumptions.